

# ANFENG XU

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## EDUCATION

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| <b>University of Southern California (USC)</b><br>Ph.D. in Electrical Engineering, Minor in Computer Science                 | Expected Dec 2026<br>GPA: 3.96/4.0 |
| <b>University of Southern California (USC)</b><br>M.S. in Electrical Engineering                                             | May 2024<br>GPA: 4.0/4.0           |
| <b>University of California, San Diego (UCSD)</b><br>B.S. in Electrical Engineering, Minor in Mathematics<br>Summa Cum Laude | June 2021<br>GPA: 3.97/4.0         |

## RESEARCH INTERESTS

Speech Language Models, Speech Processing, ASR, Large Language Models, Multimodal Modeling, Healthcare AI.

## WORK EXPERIENCE

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| <b>Meta Superintelligence Labs</b><br><i>Ph.D. Research Scientist Intern, Advisor: Zhong Meng</i>                                                                                                                          | May 2026 - Present<br><i>Menlo Park</i>     |
| Conducting research on reinforcement learning for speech language models.                                                                                                                                                  |                                             |
| <b>Meta Superintelligence Labs</b><br><i>Ph.D. Research Scientist Intern, Advisors: Yashesh Gaur, Naoyuki Kanda</i>                                                                                                        | May 2025 - Aug 2025<br><i>Seattle</i>       |
| Developed speech naturalness prediction models for conversational speech generative models (e.g., LLaMA Full-Duplex).<br>Built a large-scale synthetic data augmentation pipeline using internal speech generative models. |                                             |
| <b>Meta Reality Labs</b><br><i>Ph.D. Research Scientist Intern, Advisor: Biqiao Zhang</i>                                                                                                                                  | May 2024 - Aug 2024<br><i>Seattle</i>       |
| Conducted research to enhance the Keyword Spotting (KWS) models for wearable devices.                                                                                                                                      |                                             |
| <b>Signal Analysis and Interpretation Laboratory (SAIL), USC</b><br><i>Ph.D. Research Assistant, Advisor: Shrikanth Narayanan</i>                                                                                          | August 2021 - Present<br><i>Los Angeles</i> |
| Conduct research projects with the main focus on speech & language processing and multi-modality.<br>Lead research projects, mentor MS and junior Ph.D. students, and manage the IEMOCAP/CreativeIT data distribution.     |                                             |

## RESEARCH PROJECTS

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| <b>Automated Speech Behavior Analysis and Scoring (USC - Simons Foundation)</b>                                                                                                                                                                                                                                                                                                  | Mar 2026 – Present  |
| Developing automated methods to quantify speech behaviors in child and older adult populations with speech foundation models, LLMs, and interpretable features.                                                                                                                                                                                                                  |                     |
| <b>Low-Resource ASR and Speaker Diarization (USC - Simons Foundation)</b>                                                                                                                                                                                                                                                                                                        | Oct 2024 - Mar 2026 |
| Benchmarked speech foundation models for speaker diarization across the lifespan.<br>Developed LLM error correction methods for ASR for child-adult speech.<br>Developed an efficient ASR model with integrated speaker and timestamp prediction using serialized output training, jointly trained for speaker diarization.                                                      |                     |
| <b>Speaker Diarization for Child-Adult Interactions (USC - BU - Apple)</b>                                                                                                                                                                                                                                                                                                       | Jan 2022 - Oct 2024 |
| Proposed annotation framework and modeling pipeline to predict language capabilities of children directly from audio.<br>Developed an audio-visual child-adult speaker classification method.<br>Built a child-adult speaker diarization model with Speech Foundation Models.<br>Built simulated conversations from AudioSet to further enhance child-adult speaker diarization. |                     |

## SELECTED PUBLICATIONS

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T Feng, **A Xu**, et al. “ChildVox: A Speech, Audio, and Large Audio-Language Model Benchmark in Understanding and Characterizing Sound across Childhood,” accepted to **EMNLP** (Main), 2026.

M Nicholas, K Zareno, K Avramidis, **A Xu**, and S Narayanan. ”Speech Entrainment in Multi-Party Conversations with a Digital Agent,” **Interspeech** 2026.

**A Xu**, T Feng, S Bishop, C Lord, S Narayanan. “End-to-End Joint ASR and Speaker Role Diarization with Child-Adult Interactions,” **IEEE/ACM Transactions on Audio, Speech and Language Processing (TASLP)**, 2026.

T Feng, K Huang, **A Xu**, et al. “Voxlect: A Speech Foundation Model Benchmark for Modeling Dialects and Regional Languages Around the Globe,” **KDD**, 2026.

**A Xu\***, T Feng\*, SH Kim, S Bishop, C Lord, S Narayanan. “Large Language Models based ASR Error Correction for Child Conversations,” **Interspeech**, 2025

T Feng, **A Xu**, X Shi, S Bishop, S Narayanan. “Egocentric speaker classification in child-adult dyadic interactions: From sensing to computational modeling,” **Interspeech**, 2025

Z Shi, X Shi, **A Xu**, T Feng, H Srivastava, S Narayanan, M Mataric. “Examining Test-Time Adaptation for Personalized Child Speech Recognition,” **Interspeech**, 2025

**A Xu**, T Feng, H Tager-Flusberg, C Lord, S Narayanan. “Data Efficient Child-Adult Speaker Diarization with Simulated Conversations,” **ICASSP**, 2025.

**A Xu**, B Zhang, S Kong, Y Huang, Z Yang, S Srivastava, M Sun. “Effective Integration of KAN for Keyword Spotting,” **ICASSP**, 2025.

T Feng, **A Xu**, et al. “Can Generic LLMs Help Analyze Child-Adult Interactions Involving Children with Autism in Clinical Observation?” **NeurIPS Workshop** 2024 (GenAI for Health).

**A Xu**, K Huang, T Feng, L Shen, H Tager-Feng, S Narayanan. “Exploring Speech Foundation Models for Speaker Diarization in Child-Adult Dyadic Interactions,” **Interspeech**, 2024.

**A Xu**, K Huang, T Feng, H Tager-Feng, S Narayanan. “Audio-visual child-adult speaker classification in dyadic interactions,” **ICASSP**, 2024.

D Bose, R Hebber, T Feng, K Somandepalli, **A Xu**, S Narayanan. “MM-AU:Towards Multimodal understanding of advertisement videos,” **ACM Multimedia**, 2023.

**A Xu**, R Hebbar\*, R Lahiri\*, T Feng\*, L Butler, L Shen, H Tager-Flusberg, and S Narayanan. “Understanding Spoken Language Development of Children with ASD Using Pre-trained Speech Embeddings,” **Interspeech**, 2023.

(\* equal contribution)

## SELECTED PREPRINTS / MANUSCRIPTS UNDER REVIEW

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**A Xu**, Y Gaur, N Kanda, Z Ouyang, K Žmolíková, D Raj, S Merello, A Sun, O Kalinli. “Conversational Speech Naturalness Predictor,” arXiv:2603.01467.

**A Xu**, et al. “Speaker Role and Language Diarization for Analyzing Multilingual Interviews for Language Proficiency of Older Adults,” arXiv:2608.09032, under review for Computer Speech & Language.

T Feng, **A Xu**, J Lee, S Narayanan. “VoxCog: Towards End-to-End Multilingual Cognitive Impairment Classification through Dialectal Knowledge,” arXiv:2601.07999, under review for Computer Speech & Language.

T Feng, J Lee, **A Xu**, et al. “Vox-Profile: A Speech Foundation Model Benchmark for Characterizing Diverse Speaker Traits and Speech Properties,” arXiv:2505.14648, under review for DMLR.

## AWARDS

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MHI Ph.D. Scholar Finalist (USC, 2026)

ECE Ph.D. Screening Exam (USC, 2022): Ranked 1st for the Signals and Systems track.

Annenberg Fellowship (USC, 2021): Top-off fellowship selected among incoming Ph.D. students.

Summa Cum Laude (UCSD, 2021): GPA within the top 2% of the graduating class.

## LANGUAGES

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English (fluent), Japanese (native), Mandarin Chinese (conversational).

## SERVICES

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**Reviewer** - ICASSP, Interspeech, ACM-MM, IEEE Open Journal of Signal Processing.

**Organizer** - IEEE ASRU 2025 Satellite Workshop-AI for Children's Speech and Language.